Matrix F5 / GF(2) defaults: direct pack on; direct unpack and table reuse stay opt-in

Study `research/f5_gf2_defaults_20261017` · runs `run-20261017T0331Z` (3 rounds) and `run-20261017T0341Z` (15 rounds; F4 10 rounds) · binaries `f4_f2_bench`, `f6_ic_probe` built from `78a90d27f` · 2026-10-17

Question

Three exact switches had each passed an incremental gate on x86 CI and were left opt-in only because their protocols tied a default change to a 2× gate: `KIC_F5_DIRECT_PACK` (`research/f5_direct_pack_20260929`, 1.21×), `KIC_F5_UNPACK_DIRECT` (`research/f5_further_2x_20260929`, 1.04–1.05×), `KIC_GF2_REUSE_TABLE` (`research/gf2_table_reuse_20260929`, 1.035×). Can they be on by default without regressing on another host and on the other `gf2_elim` users?

What ran

One binary per tool (all three switches on at `78a90d27f`), arms by environment only: **on**; **nopack**, **nounpack**, **noreuse** (that switch `=0`); **off** (all three `=0`, the previous defaults). `f4_f2_bench 3 24 f5` (seven matrix-F5 cases, n ≤ 24, `Reduced` = production form and `SelectiveEchelon`, 1 and 4 threads), `f4_f2_bench f4` (matrix F4) and, in run 1, `f6_ic_probe --engines f4` (inherited F4) on three cells. Arm order rotated each round. Host: Intel Xeon @ 2.10 GHz, 4 vCPU KVM, AVX-512F, shared, unpinned, L0, load 0.36–3.23. Run 1: 84 runs; run 2: 260 runs; all exited 0. Times are the bench's per-case median of 3 repetitions (wall) and the probe's process CPU; ratios are medians over rounds (arm / on, > 1 means the defaults-on arm is faster), the range is the worst–best round pairing.

Exactness

Every F5 case (row and row-space fingerprints, rank, output terms, reduction word operations), every matrix-F4 case (basis fingerprint, word XORs) and every inherited-F4 target decision is identical in every arm and round of both runs.

Decision rule

Run 1 left direct unpack and table reuse ambiguous (on `f5_n20_m20_d4` single switches off looked faster than `on` while `off` was slower). Before run 2 the rule in `scripts/DECISION_RULE.md` was committed: a switch stays on only if in every (form, threads) cell its arm/on median is ≥ 0.97 on `f5_n20_m20_d4` and `f5_n24_m24_d4` and ≥ 1.00 on `f5_n24_m24_d4`, and, for table reuse, ≥ 0.97 on every matrix-F4 case with n ≥ 16.

Outcome (run 2)

- **Direct unpack**: fails: arm/on 0.986 on `f5_n24_m24_d4`, SelectiveEchelon, 4 threads (other cells 0.974–1.060). Stays opt-in.
- **Table reuse**: fails: arm/on 0.955 on matrix-F4 `n16_m24` (F5 cells 0.995–1.070 pass). Stays opt-in.
- **Direct pack**: on by default (run 1, 3 rounds; not repeated in run 2: arm/on 1.006–2.173 on every F5 cell; on `f5_n24_m24_d4` the worst round pairing is ≥ 1.02 in all four cells).
- Previous defaults vs all three on (off ÷ on), run 2: 1.11–1.27× on the two largest F5 cases in every cell.

Run 2, larger cases

arm	tool	form	thr	case	on ms	arm ms	arm/on	round range
noreuse	f5	form0	1	f5_n20_m20_d4	72.50	73.94	1.020	0.75–1.44
nounpack	f5	form0	1	f5_n20_m20_d4	72.50	72.08	0.994	0.77–1.57
off	f5	form0	1	f5_n20_m20_d4	72.50	82.71	1.141	0.82–1.47
noreuse	f5	form0	1	f5_n24_m24_d4	273.83	285.74	1.044	0.82–1.21
nounpack	f5	form0	1	f5_n24_m24_d4	273.83	277.97	1.015	0.84–1.28
off	f5	form0	1	f5_n24_m24_d4	273.83	315.05	1.151	0.96–1.37
noreuse	f5	form0	4	f5_n20_m20_d4	47.76	51.11	1.070	0.73–1.26
nounpack	f5	form0	4	f5_n20_m20_d4	47.76	50.11	1.049	0.73–1.27
off	f5	form0	4	f5_n20_m20_d4	47.76	55.12	1.154	0.83–1.43
noreuse	f5	form0	4	f5_n24_m24_d4	165.71	171.92	1.038	0.83–1.16
nounpack	f5	form0	4	f5_n24_m24_d4	165.71	167.08	1.008	0.84–1.17
off	f5	form0	4	f5_n24_m24_d4	165.71	183.82	1.109	0.92–1.25
noreuse	f5	form2	1	f5_n20_m20_d4	48.79	48.57	0.995	0.77–1.41
nounpack	f5	form2	1	f5_n20_m20_d4	48.79	47.54	0.974	0.79–1.36
off	f5	form2	1	f5_n20_m20_d4	48.79	55.29	1.133	0.79–1.41
noreuse	f5	form2	1	f5_n24_m24_d4	145.42	149.97	1.031	0.84–1.31
nounpack	f5	form2	1	f5_n24_m24_d4	145.42	154.11	1.060	0.85–1.23
off	f5	form2	1	f5_n24_m24_d4	145.42	183.47	1.262	1.05–1.59
noreuse	f5	form2	4	f5_n20_m20_d4	33.32	34.62	1.039	0.86–1.25
nounpack	f5	form2	4	f5_n20_m20_d4	33.32	33.03	0.991	0.80–1.26
off	f5	form2	4	f5_n20_m20_d4	33.32	42.41	1.273	1.04–1.60
noreuse	f5	form2	4	f5_n24_m24_d4	101.08	105.84	1.047	0.93–1.27
nounpack	f5	form2	4	f5_n24_m24_d4	101.08	99.66	0.986	0.87–1.18

off	f5	form2	4	f5_n24_m24_d4	101.08	122.68	1.214	1.03–1.51
noreuse	f4	n/a	1	n12_m12	7.35	7.56	1.028	0.76–1.53
noreuse	f4	n/a	1	n14_m14	24.61	25.93	1.054	0.85–1.35
noreuse	f4	n/a	1	n16_m16	298.01	293.50	0.985	0.83–1.32
noreuse	f4	n/a	1	n16_m24	77.73	74.22	0.955	0.75–1.08
noreuse	f4	n/a	1	n18_m18	1776.17	1772.28	0.998	0.92–1.08

Run 1, larger cases and the other engines

arm	tool	form	thr	case	on ms	arm ms	arm/on	round range
nopack	f5	form0	1	f5_n20_m20_d4	71.80	82.29	1.146	1.00–1.52
noreuse	f5	form0	1	f5_n20_m20_d4	71.80	70.80	0.986	0.86–1.05
nounpack	f5	form0	1	f5_n20_m20_d4	71.80	72.30	1.007	0.86–1.09
off	f5	form0	1	f5_n20_m20_d4	71.80	85.87	1.196	0.95–1.36
nopack	f5	form0	1	f5_n24_m24_d4	266.60	324.23	1.216	1.08–1.36
noreuse	f5	form0	1	f5_n24_m24_d4	266.60	285.15	1.070	0.91–1.19
nounpack	f5	form0	1	f5_n24_m24_d4	266.60	272.34	1.022	0.87–1.09
off	f5	form0	1	f5_n24_m24_d4	266.60	314.27	1.179	1.02–1.24
nopack	f5	form0	4	f5_n20_m20_d4	52.85	56.72	1.073	0.98–1.22
noreuse	f5	form0	4	f5_n20_m20_d4	52.85	46.68	0.883	0.80–0.99
nounpack	f5	form0	4	f5_n20_m20_d4	52.85	45.74	0.865	0.83–1.02
off	f5	form0	4	f5_n20_m20_d4	52.85	57.70	1.092	1.05–1.35
nopack	f5	form0	4	f5_n24_m24_d4	159.30	175.58	1.102	1.02–1.11
noreuse	f5	form0	4	f5_n24_m24_d4	159.30	157.93	0.991	0.90–1.05
nounpack	f5	form0	4	f5_n24_m24_d4	159.30	161.41	1.013	0.93–1.10
off	f5	form0	4	f5_n24_m24_d4	159.30	173.86	1.091	1.02–1.28
nopack	f5	form2	1	f5_n20_m20_d4	52.97	53.32	1.006	0.99–1.09
noreuse	f5	form2	1	f5_n20_m20_d4	52.97	48.05	0.907	0.88–0.96
nounpack	f5	form2	1	f5_n20_m20_d4	52.97	45.98	0.868	0.80–0.92
off	f5	form2	1	f5_n20_m20_d4	52.97	56.60	1.068	1.03–1.29
nopack	f5	form2	1	f5_n24_m24_d4	145.97	164.72	1.129	1.02–1.20
noreuse	f5	form2	1	f5_n24_m24_d4	145.97	153.58	1.052	0.96–1.18
nounpack	f5	form2	1	f5_n24_m24_d4	145.97	149.02	1.021	0.90–1.03
off	f5	form2	1	f5_n24_m24_d4	145.97	196.95	1.349	1.11–1.48
nopack	f5	form2	4	f5_n20_m20_d4	32.01	42.74	1.335	1.29–1.57
noreuse	f5	form2	4	f5_n20_m20_d4	32.01	33.16	1.036	0.96–1.14
nounpack	f5	form2	4	f5_n20_m20_d4	32.01	33.77	1.055	1.04–1.12
off	f5	form2	4	f5_n20_m20_d4	32.01	42.91	1.341	1.06–1.59
nopack	f5	form2	4	f5_n24_m24_d4	98.92	112.48	1.137	1.03–1.26
noreuse	f5	form2	4	f5_n24_m24_d4	98.92	105.06	1.062	0.95–1.14
nounpack	f5	form2	4	f5_n24_m24_d4	98.92	100.44	1.015	0.98–1.09
off	f5	form2	4	f5_n24_m24_d4	98.92	122.11	1.234	1.18–1.32
noreuse	f4	n/a	1	n12_m12	7.84	7.00	0.893	0.76–1.00
noreuse	f4	n/a	1	n14_m14	25.41	24.78	0.975	0.92–1.00
noreuse	f4	n/a	1	n16_m16	293.54	308.31	1.050	0.95–1.18
noreuse	f4	n/a	1	n16_m24	81.93	76.85	0.938	0.73–1.11
noreuse	f4	n/a	1	n18_m18	1793.38	1919.97	1.071	1.03–1.09
noreuse	f4	n/a	1	n20_m20	19337.74	19604.29	1.014	0.99–1.04
noreuse	f4	n/a	1	n20_m30	1478.42	1371.63	0.928	0.87–1.05

noreuse	inh	n/a	1	K0_23	2403.36	2409.95	1.003	0.87–1.10
noreuse	inh	n/a	1	K0_31	2483.26	2508.61	1.010	0.92–1.08
noreuse	inh	n/a	1	K1_23	810.36	825.74	1.019	1.00–1.04

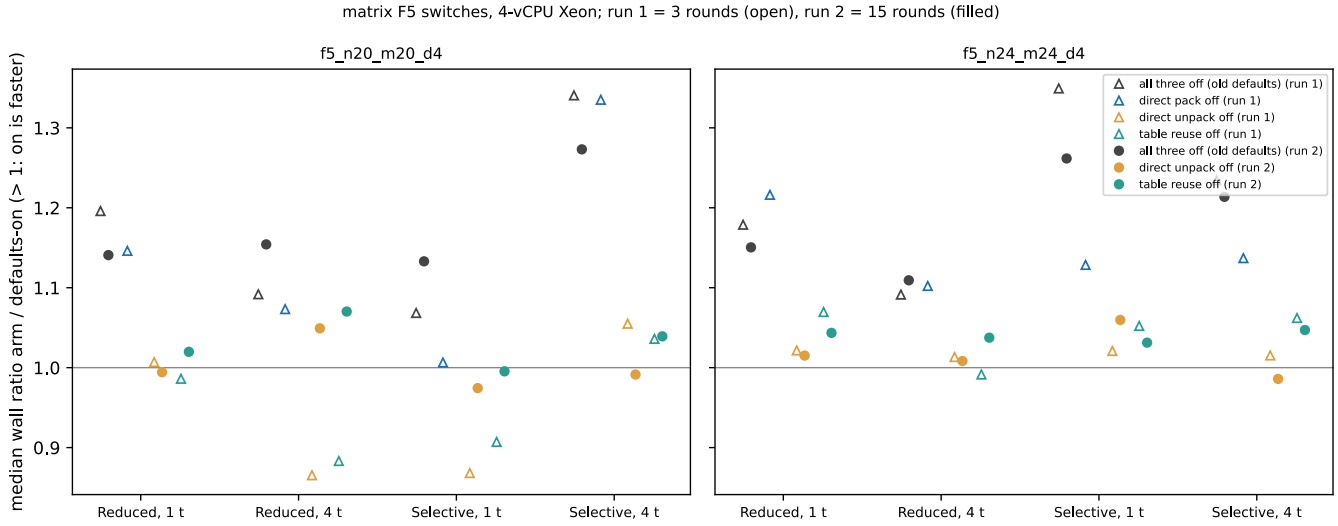


Figure 1: Median arm/on ratios on the two largest F5 cases, both runs.

Limits

One shared 4-vCPU host, L0, wall times; run-to-run ranges are wide (single runs can move ±25 %). Direct pack's measured gain here is largest on the smaller F5 cases. No IC-versus-rho statement.

Records

`reports/<run>/raw/*.jsonl` (first line: tool, arm, form, threads, round, UTC, load), `raw/*.status`, `host.txt`, `summary.txt`, `table.tsv` (`summarize.py`), `fig_f5.svg|png` (`plot.py`); scripts in `scripts/`.